Maximum-entropy null models for directed weighted networks: equations, solvers, and numerical robustness mechanisms

Technical appendix

Abstract

This appendix documents, at implementation precision, the four maximum-entropy network models implemented in the `dcms` package — DCM, DWCM, qDECM and DECM, of increasing structural richness — and the numerical machinery used to solve their defining equations on real, heavy-tailed directed networks with tens of thousands of nodes. We give the exponential-family derivation shared by all four models (Sec. 1), the per-model constraint equations (Sec. 2), the two families of fixed-point solvers and the sequence of robustness mechanisms layered on top of them — Anderson acceleration, exact hub bisection, degeneracy reduction, a stagnation/blowup recovery ladder, an argmax-streak targeted fix, and a joint block-Newton step for near-singular rows (Sec. 3) — together with the numerical evidence that motivated each one. Every equation below is transcribed from, and numerically validated against, the package's source (`dcms/models/*.py`, `dcms/solvers/*.py`).

1 Notation and the maximum-entropy framework

Let G be a directed network on N labelled nodes, represented by a binary adjacency matrix $A \in \{0,1\}^{N\times N}$ (with $A_{ii} = 0$) and, where weights are modelled, a non-negative integer weight matrix $W \in \mathbb{Z}_{\geq 0}^{N\times N}$ with $W_{ij} > 0 \iff A_{ij} = 1$. Write $k_i^{\text{out}} = \sum_{j\neq i} A_{ij}$, $k_i^{\text{in}} = \sum_{j\neq i} A_{ji}$ for the observed degree sequence and $s_i^{\text{out}} = \sum_{j\neq i} W_{ij}$, $s_i^{\text{in}} = \sum_{j\neq i} W_{ji}$ for the observed strength sequence.

All four models are instances of the same construction (Squartini & Garlaschelli, 2011)[1]: maximize the Shannon entropy $S[P] = -\sum_G P(G)\ln P(G)$ of a probability distribution P over the ensemble of directed (weighted) graphs on N nodes, subject to reproducing a chosen set of observed node-level statistics $\{C_i^{(a)}(G)\}$ in expectation, $\langle C_i^{(a)}\rangle_P = C_i^{(a)}(G^{\text{obs}})$. The constrained maximum has the exponential-family (Gibbs–Boltzmann) form

$$P(G) = \frac{e^{-H(G;\boldsymbol{\theta})}}{Z(\boldsymbol{\theta})}, \qquad H(G;\boldsymbol{\theta}) = \sum_a \sum_i \theta_i^{(a)}\, C_i^{(a)}(G), \tag{1}$$

where $\boldsymbol{\theta}$ is one Lagrange multiplier per constrained statistic per node. Because the constraints below are always separable across ordered node pairs (i, j), Z factorizes over pairs and every model reduces to solving a coupled system of $\dim(\boldsymbol{\theta})$ transcendental equations

$$\langle C_i^{(a)}\rangle_{\boldsymbol{\theta}} = C_i^{(a)}(G^{\text{obs}}), \qquad a = 1,\ldots,k,\ i = 1,\ldots,N, \tag{2}$$

in $\boldsymbol{\theta}$ — this is the root-finding problem every solver in Sec. 3 targets. Multipliers are always stored in the package as $\theta_i = -\ln x_i > 0$ (topology) and $\eta_i = -\ln\beta_i$ (weight, sign unconstrained — see Sec. 2.4), never as x_i, β_i directly, since the log-space parametrization keeps $x_i, \beta_i \in (0,1)$ automatically and conditions the Newton steps of Sec. 3.2 far better than the raw multiplicative form.

2 The four models

The four models constrain an increasing subset of $\{k^{\text{out}}, k^{\text{in}}, s^{\text{out}}, s^{\text{in}}\}$ and correspondingly increase in the number and coupling of their multipliers, summarized in Table 2 and derived individually below.

2.1 DCM — Directed Configuration Model

The DCM constrains only the degree sequences. Maximizing entropy under independent Bernoulli link variables $A_{ij} \sim \text{Bernoulli}(p_{ij})$ gives, for $i \neq j$,

$$p_{ij} = \frac{x_i y_j}{1 + x_i y_j}, \qquad x_i = e^{-\theta_{\text{out},i}},\ y_j = e^{-\theta_{\text{in},j}}, \tag{3}$$

Symbol	Meaning	Constraint(s)
$\theta_{\text{out},i}, \theta_{\text{in},i}$	degree multipliers, node i	$k_i^{\text{out}}, k_i^{\text{in}}$
$\eta_{\text{out},i}, \eta_{\text{in},i}$	strength multipliers, node i	$s_i^{\text{out}}, s_i^{\text{in}}$
$x_i = e^{-\theta_{\text{out},i}}$, $y_i = e^{-\theta_{\text{in},i}}$	topology fugacities	
$\beta_{\text{out},i} = e^{-\eta_{\text{out},i}}$, $\beta_{\text{in},i} = e^{-\eta_{\text{in},i}}$	weight fugacities	
p_{ij}	$\Pr(A_{ij} = 1)$ under the fitted model	
W_{ij} (or $\mathbb{E}[W_{ij}]$)	expected weight of arc $i \to j$	
N	number of nodes	
M	number of degeneracy-reduced groups (Sec. 3.3), $M \le N$	

Table 1: Running notation.

Model	Constrains	Unknowns	Coupled?	Source
DCM	$k^{\text{out}}, k^{\text{in}}$	$2N$: $(\theta_{\text{out}}, \theta_{\text{in}})$	—	`models/dcm.py`
DWCM	$s^{\text{out}}, s^{\text{in}}$	$2N$: $(\eta_{\text{out}}, \eta_{\text{in}})$	—	`models/dwcm.py`
qDECM	all four, two-step	$4N$: as above, solved sequentially	topology → weight only	`models/qdecm.py`
DECM	all four, jointly	$4N$: $(\theta_{\text{out}}, \theta_{\text{in}}, \eta_{\text{out}}, \eta_{\text{in}})$	fully	`models/decm.py`

Table 2: The four models, in order of increasing structural richness.

and the constraint equations (2) become

$$k_i^{\text{out}} = \sum_{j \neq i} \frac{x_i y_j}{1 + x_i y_j}, \qquad k_i^{\text{in}} = \sum_{j \neq i} \frac{x_j y_i}{1 + x_j y_i}. \tag{4}$$

The map $\theta \mapsto p_{ij}(\theta)$ is unconstrained on all of $\mathbb{R}$ (no feasibility boundary): $x_i, y_i > 0$ for any finite θ, and $p_{ij} \in (0, 1)$ automatically. This is the only one of the four models with no feasibility region to track, and it converges reliably with the plain θ-Newton iteration (Sec. 3.2) at every scale tested.

2.2 DWCM — Directed Weighted Configuration Model

The DWCM constrains only the strength sequences, under a model where each potential arc independently carries an integer weight drawn from a geometric distribution with success parameter $1 - z_{ij}$, $z_{ij} = \beta_{\text{out},i}\beta_{\text{in},j}$. The resulting expected weight and constraint equations are

$$\mathbb{E}[W_{ij}] = \frac{z_{ij}}{1 - z_{ij}}, \qquad s_i^{\text{out}} = \sum_{j \neq i} \frac{\beta_{\text{out},i}\beta_{\text{in},j}}{1 - \beta_{\text{out},i}\beta_{\text{in},j}}, \qquad s_i^{\text{in}} = \sum_{j \neq i} \frac{\beta_{\text{out},j}\beta_{\text{in},i}}{1 - \beta_{\text{out},j}\beta_{\text{in},i}}. \tag{5}$$

Feasibility: $z_{ij} < 1$ for every realizable pair $i \neq j$, i.e. $\eta_{\text{out},i} + \eta_{\text{in},j} > 0$ in log-space. Unlike DCM, this is a genuine constraint: as any pair approaches $z_{ij} \to 1^-$, both $\mathbb{E}[W_{ij}]$ and the Jacobian entries diverge — the source of the "hub" numerical pathology addressed in Sec. 3.4.

2.3 qDECM — Quasi-DECM (two-step, approximate)

The qDECM constrains all four sequences but keeps the topology and weight sub-problems *decoupled*: the link probability p_{ij} is fixed by a plain DCM fit (Sec. 2.1) and only then does a second, *conditioned* weight step introduce β:

$$s_i^{\text{out}} = \sum_{j \neq i} \frac{p_{ij}}{1 - \beta_{\text{out},i}\beta_{\text{in},j}}, \qquad s_i^{\text{in}} = \sum_{j \neq i} \frac{p_{ji}}{1 - \beta_{\text{out},j}\beta_{\text{in},i}}, \tag{6}$$

with p_{ij} from the already-solved DCM held fixed throughout. The conditional expected weight given a link, $\mathbb{E}[W_{ij} \mid A_{ij} = 1] = 1/(1 - z_{ij}) =: G_{ij}$, is the same geometric factor as DWCM's, so $\mathbb{E}[W_{ij}] = p_{ij}\, G_{ij}$ (numerator p_{ij}, *not* $p_{ij}z_{ij}$ — a frequent sign error this appendix flags explicitly since it silently produces a self-consistent but wrong fixed point). Separability is the entire point of this construction: because p_{ij} does not depend on β, the weight step (Eq. 6) is itself exactly DWCM's fixed-point structure (Eq. 5) with $\beta\beta$ replaced by p_{ij} in the numerator, so every DWCM solver

and robustness mechanism (Secs. 3.1–3.4) applies unmodified. This makes qDECM the cheapest of the three weighted models to solve robustly, at the cost of being an *approximation*: it does not reproduce the exact maximum-entropy ensemble under the joint (degree, strength) constraint set, only a product of two separately-exact ensembles.

2.4 DECM — Directed Enhanced Configuration Model (exact, fully coupled)

The DECM is the exact maximum-entropy model under the joint constraint set: unlike qDECM, the weight multipliers enter the link probability itself, so the degree and strength equations are *coupled* through a single partition function per pair. For $i \neq j$, with $z_{ij} = \eta_{\text{out},i} + \eta_{\text{in},j}$,

$$p_{ij} = \sigma\big(-\theta_{\text{out},i} - \theta_{\text{in},j} - \ln \operatorname{expm1}(z_{ij})\big), \qquad G_{ij} = \frac{1}{1 - e^{-z_{ij}}} = -\frac{1}{\operatorname{expm1}(-z_{ij})}, \tag{7}$$

where $\sigma(\cdot)$ is the logistic sigmoid and $\operatorname{expm1}(x) = e^x - 1$ is used throughout in place of $e^x - 1$ for numerical stability near $x = 0$ (the region every feasibility floor in Sec. 3.5 exists to protect). The expected weight is again $\mathbb{E}[W_{ij}] = p_{ij} G_{ij}$, and the $4N$ coupled constraint equations are

$$k_i^{\text{out}} = \sum_{j \neq i} p_{ij}, \quad k_i^{\text{in}} = \sum_{j \neq i} p_{ji}, \quad s_i^{\text{out}} = \sum_{j \neq i} p_{ij} G_{ij}, \quad s_i^{\text{in}} = \sum_{j \neq i} p_{ji} G_{ji}. \tag{8}$$

Feasibility: $z_{ij} = \eta_{\text{out},i} + \eta_{\text{in},j} > 0$ for every realizable pair $i \neq j$ — note this is a constraint on the *pairwise sum*, not on the individual multipliers: an individual $\eta_{\text{out},i}$ may be negative provided every pair it participates in still sums positive (Sec. 3.4, "negative-η relaxation"). Because θ and η both enter p_{ij} (Eq. 7), the Jacobian of Eq. 8 is not block-diagonal in (θ, η): this coupling, absent from qDECM by construction, is both what makes DECM the exact model and the entire reason Secs. 3.4–3.7 exist.

3 Numerical methods

All four models reduce to the root-finding problem of Eq. 2; the package solves it with a single iteration, a θ-space Newton step (Sec. 3.2) accelerated by Anderson mixing (Sec. 3.1), run on the degeneracy-reduced system (Sec. 3.3), and — for the harder models, principally DECM — five additional mechanisms (Secs. 3.4–3.7) that were each added in response to a specific, reproducible failure mode on real heavy-tailed networks. Every solver returns the best (lowest residual) iterate seen over the run, not merely the final one, and reports convergence by the ℓ_∞ *relative* residual

$$\text{MRE}(\theta) = \max_{a,i:\, C_i^{(a)}(G^{\text{obs}}) > 0} \frac{\left|\langle C_i^{(a)} \rangle_\theta - C_i^{(a)}(G^{\text{obs}})\right|}{C_i^{(a)}(G^{\text{obs}})}, \tag{9}$$

which is the acceptance criterion throughout this appendix (MRE $\leq$ tol, default tol $= 10^{-5}$ for the validation runs of Sec. 5).

3.1 Anderson acceleration

Let $g(\theta) = \theta + \Delta\theta$ be one Newton step of Sec. 3.2. Given the last m residuals $r_k = g(\theta_k) - \theta_k$, Anderson mixing finds the affine combination of the last m map outputs that minimizes the mixed residual's norm,

$$\min_{c \in \mathbb{R}^m,\ \sum_k c_k = 1} \Big\| \sum_k c_k\, r_k \Big\|^2, \tag{10}$$

a small $m \times m$ least-squares problem solved every iteration at $O(m^2)$ cost, turning the linear-rate fixed-point map into an effectively super-linear one; the package uses $m = 10$ by default, reducing iteration counts by a measured 5–50× on well-conditioned synthetic instances (Sec. 5). Two guards protect the mixing from corrupting the run: (i) residuals are entered row-normalized by their own component-wise magnitude before the least-squares solve, so a handful of large-residual hub components cannot dominate the fitted coefficients; (ii) if a mixed iterate's residual exceeds `blowup_factor` times the best residual seen so far, the mixing history is discarded and the run falls back to a plain (un-mixed) step for that iteration — the first line of defence against the divergence modes that motivate Sec. 3.5.

3.2 θ-Newton: coordinate-wise Newton step

Rather than solving $\theta = g(\theta)$, treat Eq. 2 as a root-finding problem $F(\theta) = 0$, $F_i^{(a)}(\theta) := \langle C_i^{(a)} \rangle_\theta - C_i^{(a)}(G^{\text{obs}})$, and take one *diagonal* (coordinate-wise) Newton step per multiplier, holding every other coordinate fixed:

$$\Delta\theta_i^{(a)} = -F_i^{(a)}(\theta) \Big/ \frac{\partial F_i^{(a)}}{\partial \theta_i^{(a)}}(\theta), \tag{11}$$

clipped to $|\Delta\theta_i^{(a)}| \le$ `max_step` to bound the step size near hubs, and applied Gauss–Seidel style (fresh values feed forward within the same sweep) with the out-multipliers updated first. For the DCM's out-degree equation the diagonal Jacobian entry is $\partial F_i^{\text{out}}/\partial\theta_{\text{out},i} = -\sum_{j\neq i} p_{ij}(1-p_{ij})$, always strictly negative, giving a step whose numerator (residual) and denominator (local curvature, roughly the node's degree) both scale with the node's connectivity — this is the mechanism by which hub nodes, which break FP-GS's $\rho < 1$ assumption, are handled gracefully by Newton instead: a large residual is matched by a large curvature, keeping the step well-scaled. For the weighted models the analogous diagonal is $\partial F_i^{\text{out}}/\partial\eta_{\text{out},i} = -\sum_{j\neq i} p_{ij}G_{ij}(G_{ij}-1) - p_{ij}(1-p_{ij})G_{ij}^2$ (DECM; qDECM's is the first term alone, since p_{ij} does not depend on η there), always strictly negative by the same argument, guaranteeing descent. DECM alternates two coupled Gauss–Seidel–Newton passes per iteration: pass 1 updates $(\theta_{\text{out}}, \eta_{\text{out}})$ from row sums using the *input* $(\theta_{\text{in}}, \eta_{\text{in}})$; pass 2 updates $(\theta_{\text{in}}, \eta_{\text{in}})$ from column sums using the pass-1 output — the residual reported for the iteration is always evaluated at the pass's *input* θ, so it reflects the previous iteration's correction honestly, never an artificially zeroed in-progress value. Anderson acceleration (Sec. 3.1) applies identically on top of this map. This is the package's only iteration, used for every model at every scale.

The z-floor. Whenever a model's equations contain a term of the form $1/(1-z_{ij})$ or $1/\text{expm1}(\pm z_{ij})$ (DWCM, qDECM's weight step, DECM), the raw Newton step in Eq. 11 can propose a $z_{ij} \to 0^+$ (or, for DECM, $z_{ij} \le 0$) that violates feasibility and sends the corresponding term to $+\infty$. Every such solver therefore computes, after each proposed step, the tightest pair sum a node participates in and applies a trust-region-style line search: the step is scaled back (never reversed) so that the tightest pair never drops below $\max(\texttt{_Z_NEWTON_FRAC} \times z_{\min},\ \texttt{z_clamp})$ — i.e. at most a fractional (default 50%) shrinkage of the current tightest margin per step, with a hard floor `z_clamp` below which the corresponding weight factor is simply capped rather than allowed to diverge numerically. `z_clamp` is a genuine per-model tuning constant, not a universal one: 10^{-6} for DECM, 10^{-8} for qDECM, 10^{-15} for DWCM in the current release — each independently selected for its own model's numerics, with no evidence a single shared value would suit all three (Sec. 4.1).

3.3 Degeneracy reduction

Proposition 1 (Label-swap symmetry)**.** *If two nodes i, j of a given model share identical sufficient statistics — $(k_i^{\text{out}}, k_i^{\text{in}}) = (k_j^{\text{out}}, k_j^{\text{in}})$ for DCM; $(s_i^{\text{out}}, s_i^{\text{in}}) = (s_j^{\text{out}}, s_j^{\text{in}})$ for DWCM; the full 4-tuple for qDECM's weight step and for DECM — then at the (unique, by strict concavity of the log-likelihood) maximum-entropy solution their multipliers coincide.*

Proof sketch. Relabelling $i \leftrightarrow j$ is a symmetry of the constraint set (their target statistics are equal, so the relabelled problem is identical to the original) and hence of the log-likelihood $\ell(\theta) = -H(G^{\text{obs}};\theta) - \ln Z(\theta)$, which is invariant under simultaneously swapping $\theta_i \leftrightarrow \theta_j$. Since ℓ is strictly concave in θ (a standard property of this exponential family), its maximizer is unique, and a symmetry of a function with a unique maximizer must fix that maximizer — hence $\theta_i = \theta_j$ at the solution. □ □

On real, heavy-tailed degree/strength sequences a large fraction of nodes — typically the low-degree bulk — share exact sufficient-statistics tuples, so the number of distinct groups M is often far smaller than N. Proposition 1 lets every solver above be rewritten in terms of M group-level unknowns, replacing the dominant $O(N^2)$ pairwise cost of every step with $O(M^2)$ (each group g of size $\text{mult}[g]$ contributes $\text{mult}[g]$ times to every sum it enters, with one term removed for the group's own diagonal self-interaction to preserve $i \neq j$ exactly). This is a mathematically exact reformulation, not an approximation, and is the default execution path for every model in the package once expanded back to per-node shape, measured at 10–67× wall-clock speedups on real social-network data (Table 3).

Remark 1. *All four models carry a genuine additive gauge freedom on top of Proposition 1: because every model's equations depend on θ only through the sum $\theta_{\text{out},i}+\theta_{\text{in},j}$ (and, where a strength part exists, separately through $\eta_{\text{out},i}+\eta_{\text{in},j}$), the shift $\theta_{\text{out}} \mapsto \theta_{\text{out}} + c$, $\theta_{\text{in}} \mapsto \theta_{\text{in}} - c$ leaves every p_{ij} (and, for the analogous η-shift, every W_{ij}) exactly unchanged. DECM has two independent such directions. Consequently two solutions of the same instance are not expected to land*

Model	Network	$N \to M$	Speedup
DCM	social network A	$1304 \to 61$	24.2×
DCM	social network B	$20914 \to 2310$	43.5×
DWCM	social network C	$15168 \to 1860$	47.1×
DWCM	social network D	$31874 \to 3083$	66.8×
qDECM (weight step)	social network B	$20914 \to 4070$	23.5×
DECM	social network C	$15168 \to 3003$	$\sim$ 10–25×

Table 3: Degeneracy-reduction speedups measured on real (anonymized) directed social networks; `tol`$\approx 10^{-9}$–10^{-10}, θ-Newton, Anderson depth 10.

on identical raw θ even when both are converged to machine precision; any comparison between two solutions of the same instance must be made through a gauge-invariant quantity (p_{ij} and/or $\mathbb{E}[W_{ij}]$), never raw θ.

3.4 Exact hub bisection

A node i with strength-to-degree ratio $s_i/k_i \gg 1$ forces its weight fugacity toward the feasibility boundary ($\beta_i \to 1$ for qDECM, $\eta_i \to 0$ for DECM) at the true solution, where the Jacobian entries of Sec. 3.2 diverge and the coordinate Newton step becomes numerically unstable even though a solution provably exists (the boundary is approached, never reached, for any finite observed strength). For nodes flagged as hubs ($s_i/\hat{k}_i >$ `hub_sk_threshold`, where $\hat{k}_i = \sum_j p_{ij}$ is the *expected* degree from the current topology fit), the package instead finds that node's weight multiplier *exactly* each iteration by 1-D bisection on the corresponding constraint equation, holding every other multiplier fixed:

$$f(\beta_{\text{out},i}) = \sum_j \frac{p_{ij}}{1 - \beta_{\text{out},i}\beta_{\text{in},j}} - s_i^{\text{out}} = 0 \quad \text{(qDECM)}, \qquad f(\eta_{\text{out},i}) = \sum_j p_{ij}(\eta)\, G_{ij}(\eta) - s_i^{\text{out}} = 0 \quad \text{(DECM)}, \tag{12}$$

each f monotone in its argument (increasing in β, decreasing in η) on the feasible interval, guaranteeing a unique root found to machine precision in ~ 60 bisection halvings. Because a node can be simultaneously an out-hub and an in-hub, all out-hubs are bisected first against the current in-side values, then all in-hubs against the fresh out-side values, for `hub_bisect_max_sweeps` (default 3) total Gauss–Seidel passes to reach mutual consistency; Anderson mixing (Sec. 3.1) excludes hub components from its residual history and overwrites them with the bisection value after every mixed step, so hub multipliers are never corrupted by an extrapolated mixing coefficient.

Negative-η relaxation (DECM only). DECM's feasibility constraint is on the pairwise sum $\eta_{\text{out},i} + \eta_{\text{in},j} > 0$ (Sec. 2.4), not on $\eta_{\text{out},i} \geq 0$ individually; an extreme-outlier hub can require $\eta_{\text{out},i} < 0$ while every pair it participates in still sums positive. The bisection bracket is widened to allow this for nodes proven "unreachable" under the naive $\eta \geq 0$-only rule, with the growth of the resulting weight factor G_{ij} capped at a relative 5% per bisection call (converted back to an η floor via G's exact inverse) to prevent a single relaxed hub's update from destabilizing its tightest partner's own, undamped Newton step on the same iteration.

3.5 DECM stagnation and blowup recovery

Even with Secs. 3.1–3.4, some hub-heavy DECM instances stall at a *quasi-fixed point*: the deterministic Newton/Anderson dynamics reproduce the same trajectory indefinitely, so a plain rollback-and-retry cannot escape it. Two triggers share one escalation ladder:

- **Stagnation**: no record (ℓ_∞-best) improvement for `patience` consecutive iterations (or, with `patience_rel_tol`>0, no improvement exceeding that *relative* fraction — guarding against a record that creeps down by a negligible amount every iteration without ever formally stalling).
- **Repeated blowup**: an isolated Anderson blowup (Sec. 3.1) is left to its cheap rollback; only if it recurs with no record improvement in between does it escalate to the same ladder.

The ladder itself has three tiers, each attempted only after the previous one has also failed to beat the record: (1) a free soft reset (clear Anderson history, one plain Newton step from the best iterate, no noise); (2) an optional switch to the joint block-Newton step (`stall_block_newton_gate`); (3) a perturbed restart from the best iterate with multiplicative, log-scale Gaussian noise, $\theta_i \mathrel{*}= \exp \mathcal{N}(0, \sigma)$, σ starting at `noise_base` and doubling on each further failed restart up to a cap, resetting only on a genuine improvement — multiplicative rather than additive noise because a fixed absolute kick is simultaneously too weak to move a topological hub sitting near its θ boundary and catastrophic for a strength hub

sitting near its η floor. The run gives up (converged = False) only after `max_stalls` restarts *at the noise cap* in a row without improving the record. On a validation instance ($N = 15168$, $M = 3003$ after reduction, several extreme-hub nodes), this ladder converged an otherwise-stalled run unattended in 9460 iterations (one repeated-blowup restart, four stagnation restarts) to MRE = 9.45×10^{-6}.

3.6 Argmax-streak targeted fix

On a subset of real instances, once the rest of the network is well below tolerance, a single (node, side) pair's equations remain the residual's argmax for many consecutive iterations: the global step still corrects it, but too slowly relative to everything else. Tracking the identity of the argmax across iterations, once the same pair has held it for `streak_fix_threshold` consecutive iterations, that pair's (θ, η) is solved directly by nested bisection — the k-equation for θ and the s-equation for η, at *fixed* $\varphi = \theta + \eta$ rather than alternating with one held fixed at a time (an earlier θ-fixed/η-fixed alternation failed to converge whenever η was large, the two 1-D searches fighting each other along what is effectively a single direction once φ is introduced), including the negative-η bracket of Sec. 3.4 where applicable and both sides (correcting one side can shift the other's own balance).

3.7 Joint block-Newton step for near-singular rows

A further plateau shape survives every mechanism above: a family of nodes with $s_i \approx k_i$, i.e. $G_{ij} \approx 1$ for their dominant partners. Sec. 3.2's Newton step treats θ (from the k-equation) and η (from the s-equation) as *decoupled* scalar unknowns, ignoring the equations' cross-derivative — but as $G \to 1$ the two equations become nearly the same function of (θ, η), so the row's true 2×2 Hessian block is nearly singular and the decoupled iteration crawls.

Derivation. Write, per group/node and summed over its partners with multiplicity m_j (Sec. 3.3), $p = p_{ij}$, $q = 1 - p$, $G = G_{ij}$, $g := G - 1$ (≥ 0), and define

$$A = \textstyle\sum_j m_j\, pq, \qquad B_p = \sum_j m_j\, pq\, g, \qquad C_p = \sum_j m_j\, pq\, g^2, \qquad E = \sum_j m_j\, p\, G\, g. \tag{13}$$

A is exactly Sec. 3.2's θ-diagonal $-\partial F_k/\partial\theta$; the row's exact 2×2 Hessian block of (F_k, F_s) with respect to (θ, η) is $\left(\begin{smallmatrix} A & B \\ B & C \end{smallmatrix}\right)$ with $B = A + B_p$ and $C = A + 2B_p + C_p + E$ (the latter is Sec. 3.2's η-diagonal, re-expressed in the g-basis). A direct expansion gives the determinant in a cancellation-safe form,

$$\det = AC - B^2 = A\,E + \left(A\,C_p - B_p^2\right), \tag{14}$$

where the bracketed term is non-negative by the Cauchy–Schwarz inequality applied to the sequences $\sqrt{m_j pq}$ and $\sqrt{m_j pq}\, g_j$, and $A, E \geq 0$ outright ($p, G, g \geq 0$) — so $\det \geq 0$ always, as it must for a principal minor of a concave log-likelihood's (negative semi-definite) Hessian, and Eq. 14 avoids ever forming the much larger, near-cancelling quantities AC and B^2 directly (both $O(A^2)$ when $B_p \ll A$, while their difference is $O(A \cdot E)$, orders of magnitude smaller on the near-singular rows this mechanism targets). The relative conditioning $\det/(AC) = 1 - \rho^2$ (with $\rho = B/\sqrt{AC}$ the row's correlation coefficient between the two equations) was measured at 10^{-4}–10^{-5} on the affected rows of two real instances, meaning the plain decoupled iteration of Sec. 3.2 contracts by only $\rho \approx 1 - 10^{-5}$ per step along that row's own soft direction — thousands of iterations per e-fold of progress.

Mechanism (`block_newton_gate`). For a row with $\det/(AC) <$ `block_newton_gate`, solve the exact joint step $(\Delta\theta, \Delta\eta) = \left(\begin{smallmatrix} A & B \\ B & C \end{smallmatrix}\right)^{-1}(F_k, F_s)$ instead of the two decoupled scalar steps, jointly rescaled to respect `max_step` and the z-floor of Sec. 3.2. `block_newton_gate`= 0 (the default) reproduces the decoupled step bit-for-bit; hub-owned rows (Sec. 3.4) are always excluded.

The runaway-row failure mode and `block_newton_min_rel`. A row with $s_i = k_i$ *exactly* has no finite solution: its fixed point lies at $\theta \to -\infty, \eta \to +\infty$ along the soft direction $(\theta - c, \eta + c)$ (the direction Eq. 14's near-zero eigenvalue corresponds to), with the residual only ever approaching zero in the limit. Once such a row's residual is already small, the joint step of the previous paragraph has nothing stopping it from continuing to march that row along the soft direction at `max_step` per iteration indefinitely, well past any useful precision — and several rows doing this simultaneously destabilizes their neighbours through the shared partition function. `block_newton_min_rel` (default $0.1 \times$ `tol` whenever the gate is active) excludes a row from the joint step once its own relative residual is already below that value, leaving it to the (slow, but harmless) decoupled step from then on.

Remark 2 (Validated scope)**.** `block_newton_gate` *alone, without* `block_newton_min_rel`*, is* not *a safe default: on a stability check re-solving 14 already-converged real instances for 30 further iterations, a bare gate made 4 of them*

measurably worse (one instance's residual rose transiently to $O(10^3)$ above tolerance before settling; another had not recovered by the 30th iteration) and diverged a fifth, previously easy, instance entirely when solved from a fresh initial condition. Paired with `block_newton_min_rel`*, the mechanism closed the two hardest instances of the validation set (Sec. 5) from long-standing plateaus in under 100 iterations each, but the full stability check above has not yet been repeated with* `block_newton_min_rel` *active. It is documented and shipped as an opt-in, per-instance mechanism, not (yet) a default recipe.*

4 Implementation notes

Dense/chunked execution. For N (or, on the degeneracy-reduced path, M) below a per-model threshold (5000 for DCM/DWCM, 2000 for qDECM/DECM), every solver materializes the full $N \times N$ (or $M \times M$) probability/weight matrix once per iteration; above it, rows are processed in chunks of 512 to keep peak memory at $O(\text{chunk} \times N)$ rather than $O(N^2)$, with results proven identical to the dense path up to floating-point summation order.

GPU/device. All four models' degeneracy-reduced solvers accept a `device` string (`"cpu"` default float64, or `"mps"`/`"cuda"` float32 — MPS has no float64 support at all, and CUDA float64 throughput makes float32 the practical choice there too); the returned result is always converted back to CPU/float64 regardless. DECM's stagnation-recovery machinery (Sec. 3.5) stays internally CPU-only by design and interoperates transparently with a non-CPU main loop.

4.1 On `z_clamp` consistency across models

Sec. 3.2 noted that `z_clamp` is model-specific by design (10^{-6} DECM, 10^{-8} qDECM, 10^{-15} DWCM). As of this writing all three values are independently tunable parameters, each tied (where applicable) to the same value the corresponding model's own residual/likelihood computation uses internally, so the equations a solver converges and the residual it is judged against never silently disagree. DECM's value received a dedicated multi-instance investigation before being fixed at 10^{-6}; qDECM's and DWCM's have not, and are exposed for parity and debuggability rather than because a different value is known to help either model — both already converge their full real-network validation set at the shipped defaults.

5 Validation

Synthetic benchmarks. On synthetic power-law networks ($N \in \{1000, 5000\}$, multiple random seeds, Chung–Lu topology with Pareto-distributed target degrees/strengths), θ-Newton Anderson(10) reaches MRE $\leq 10^{-5}$ on 100% of tested seeds for all four models; DECM additionally falls back to a `"qdecm"` warm-start and then a random restart on any seed the plain `"degrees"` initial condition fails to converge from, resolving the remainder.

Real-network validation. The package's target real-world validation set — 19 directed, weighted networks drawn from online social media data, N ranging from a few hundred to $\sim 10^5$ nodes, several with extreme strength-to-degree hubs (s/k up to ~ 150) — is fully solved by DECM (the exact, most demanding model) to MRE $\leq 10^{-5}$ using the mechanisms of this appendix; DWCM and qDECM, structurally easier per Sec. 2.3, converge the same set with Secs. 3.1–3.4 alone. The two hardest instances in the set required Sec. 3.7's mechanism specifically: one converged from a 3.52×10^{-5} plateau in 46 iterations with `block_newton_gate` alone; the other, whose near-singular-row family was larger and triggered the runaway-row failure mode of Sec. 3.7, converged from a 1.85×10^{-4} plateau in 94 iterations once `block_newton_min_rel` was added.

Data and code availability

The `dcms` package, including every equation and mechanism described in this appendix, is available at the repository this appendix ships with; `README.md` §§2.1–2.9 give the equivalent documentation in narrative/API-reference form, and `tests/` gives executable, machine-checked specifications of every claim above (constraint equations solved to a known synthetic ground truth; the block-Newton step verified against a finite-difference Newton solve to 1.6×10^{-7} relative error; the degeneracy reduction verified, at random points and with groups of multiplicity > 1, to give a group-level residual identical to the per-node residual of the expanded parameters, and the expanded solutions to satisfy the original per-node equations).

References

[1] Squartini, T. & Garlaschelli, D. Analytical maximum-likelihood method to detect patterns in real networks. *New Journal of Physics* **13**, 083001 (2011).

[2] Walker, H. F. & Ni, P. Anderson acceleration for fixed-point iterations. *SIAM Journal on Numerical Analysis* **49**, 1715–1735 (2011).

[3] Kelley, C. T. *Iterative Methods for Linear and Nonlinear Equations* (SIAM, 1995).

[4] Vallarano, N. *et al.* Fast and scalable likelihood maximization for exponential random graph models with local constraints. *Scientific Reports* **11**, 15227 (2021).